

Gray Matter Damage and Long-Term Disability, Cognitive Outcome, and Survival in Patients With Multiple Sclerosis

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Abstract

Background and Objectives

Gray matter (GM) damage is a key determinant of clinical disability in multiple sclerosis (MS), but its relevance to very long-term outcomes remains insufficiently characterized. We assessed associations between baseline and 12-month MRI measures of GM and white matter (WM) damage and long-term outcomes in relapse-onset MS.

Methods

We conducted a prospective longitudinal single-center cohort study of relapse-onset MS patients enrolled between 1993 and 1998. Participants underwent clinical and 1.5T brain MRI assessments at baseline and after 12 months. MRI measures included T2-lesion and T1-lesion volumes, GM, WM, and thalamic volume fractions, magnetization transfer ratio (MTR) of GM, thalamus, normal-appearing WM, and WM lesions. Escalation to high-efficacy disease-modifying therapies (HE-DMTs) was recorded. Outcomes included Expanded Disability Status Scale (EDSS) worsening, evolution to a more severe disease stage, MS-related death, and cognitive deterioration. Associations were evaluated using LASSO-regularized logistic regression with internal bootstrap validation.

Results

Seventy-three relapse-onset MS patients (mean age = 33.0 years; female = 69.9%) were followed for a median of 25.9 years (interquartile range = 18.1–27.1; patients lost during follow-up = 11). EDSS worsening occurred in 79.5% of patients, 72.6% evolved to a more severe disease stage, 24.7% died from MS-related causes, and 39.5% of patients with cognitive data (17/43) experienced cognitive deterioration. EDSS worsening was associated with lower baseline GM fraction (GMF) (standardized- β = -0.561), escalation to HE-DMTs (standardized- β = 0.059), lower baseline thalamic fraction (standardized- β = -0.115), lower baseline normal-appearing WM MTR histogram peak height (standardized- β = -0.010), and greater 12-month decline in GM MTR histogram peak height (standardized- β = -0.115) (area under the curve [AUC] = 0.843, 95% CI 0.744–0.941). Evolution to a more severe disease stage was associated with higher 12-month EDSS score change (standardized- β = 0.167) and lower baseline GMF (standardized- β = -0.151) (AUC = 0.793, 95% CI 0.670–0.916). MS-related death was associated with male sex (standardized- β = 0.063), higher baseline EDSS score (standardized- β = 0.182), lower baseline GMF (standardized- β = -0.304), lower baseline GM MTR histogram peak height (standardized- β = -0.050), and greater 12-month decline in mean thalamic MTR (standardized- β = -0.370) (AUC = 0.887, 95% CI 0.813–0.961). Cognitive deterioration was associated with older baseline age (standardized- β = 0.197), lower baseline brain parenchymal fraction (standardized- β = -0.039), and lower baseline mean GM MTR (standardized- β = -0.376) (AUC = 0.891, 95% CI 0.788–0.994).

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Supplementary Material

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Glossary

AUC = area under the curve; **BPF** = brain parenchymal fraction; **BRB-N** = Brief Repeatable Battery of Neuropsychological Tests; **CIS** = clinically isolated syndrome; **EDSS** = Expanded Disability Status Scale; **FOV** = field of view; **GM** = gray matter; **HE-DMT** = high-efficacy disease-modifying therapy; **IQR** = interquartile range; **LV** = lesion volume; **MS** = multiple sclerosis; **MTR** = magnetization transfer ratio; **RR** = relapsing-remitting; **SP** = secondary progressive; **TE** = echo time; **TR** = repetition time; **WM** = white matter.

Discussion

Brain GM damage showed consistent associations with disability worsening and evolution, mortality, and cognitive deterioration over 26 years in relapse-onset MS, supporting the long-term clinical relevance of GM-focused MRI markers.

Introduction

MRI is a cornerstone of multiple sclerosis (MS) diagnosis, owing to its sensitivity in detecting demyelinating lesions and its ability to assess underlying disease mechanisms.^{1,2} In addition to its diagnostic utility, MRI also offers important prognostic insights, particularly during early disease stages.²⁻⁹

In patients presenting with clinically isolated syndrome (CIS), white matter (WM) lesion count, volume, and distribution, the presence of gadolinium-enhancing lesions and accrual of new lesions within 2–5 years are strongly associated with conversion to clinically definite MS and disability progression.²⁻⁹ However, in patients with established MS, particularly those with long-standing disease, the association between conventional lesion-based MRI measures and clinical outcomes becomes weaker.⁶

This underscores the need for imaging biomarkers that more directly reflect neurodegenerative processes, including demyelination and neuroaxonal loss, which are strongly related to irreversible disability.⁶ Advanced MRI techniques,¹⁰ including magnetization transfer ratio (MTR) imaging,^{10,11} which reflects myelin density and axonal integrity, and brain atrophy, particularly gray matter (GM) loss,^{6,12,13} have emerged as sensitive and specific tools to detect diffuse tissue damage and neurodegeneration.¹ GM atrophy and MTR reductions have shown stronger associations with physical disability and cognitive impairment than conventional MRI measures.^{6,10-13} Moreover, these advanced MRI markers seem to be better associated with long-term progression in patients with definite MS, although most available studies are limited to follow-up durations of 5–12 years.^{6,10-15} At present, the associations of GM damage with clinical outcomes over decades remains insufficiently characterized.

The increasing use of high-efficacy (HE) disease-modifying therapies (DMTs) and accumulating evidence supporting their early use initiation¹⁶ highlight the need to identify patients at greatest risk of poor outcomes early in the disease course. In this context, long-term prospective studies employing multiparametric MRI assessments are essential.

At our institution, a longitudinal observational study of patients with relapse-onset MS has been ongoing for more than 2 decades, with standardized clinical and brain MRI assessments performed at baseline and after 12 months, and subsequent clinical follow-ups at 4.5,¹⁷ 8,¹⁸ and 13 years.¹⁹ Earlier analyses of this cohort suggested an increasing prognostic relevance of GM damage over time that outweigh that of WM. However, the role of GM MRI measures in relation to very long-term clinical outcomes remains insufficiently characterized.

To further explore this hypothesis, here we present the results from the 26-year clinical follow-up of this cohort. This uniquely long observational period allowed us to investigate whether MRI variables are associated not only with disability worsening and cognitive deterioration but also with MS-related death, arguably the most definitive outcome of disease progression.²⁰⁻²²

Methods

Standard Protocol Approvals, Registrations, and Patient Consents

Approval was received from the ethical standards committee on human experimentation of IRCCS Ospedale San Raffaele (Protocol N° 2009-74). Written informed consent was obtained from all patients before study participation in accordance with the Declaration of Helsinki.

Study Population

Patients included had clinically definite MS²³ for ≥ 2 years, with a relapsing-remitting (RR) or secondary progressive (SP) disease phenotype,^{24,25} or CIS suggestive of MS, with the first clinical attack in the preceding 3 months and ≥ 4 T2-hyperintense WM lesions.¹⁷ All patients had participated in earlier follow-up studies^{17-19,26} and were consecutively enrolled in this long-term observational investigation.

Clinical Assessment

Neurologic assessment, EDSS²⁷ scoring, clinical phenotype classification, as well as documentation of ongoing DMT use were performed at baseline and at prespecified follow-up

visits.¹⁷⁻¹⁹ Clinical status was assessed by in-person neurologic examination. When not available/feasible, EDSS was obtained by structured telephone interview using a validated telephone-based EDSS approach.²⁸ Disease stage at follow-up (CIS, RRMS, SPMS, or MS-related death) was determined using all available clinical information, including EDSS score (derived by telephone when applicable), longitudinal history of disability progression, relapse history, review of medical records, and careful structured interviews with patients and caregivers.

Escalation from moderate-efficacy (interferon beta-1a, glatiramer acetate, azathioprine, or methotrexate) to high-efficacy DMTs (mitoxantrone, cyclophosphamide, fingolimod, siponimod, natalizumab, rituximab, or ocrelizumab) was recorded longitudinally whenever treatment changes occurred throughout the follow-up period.

Disability worsening was defined as an EDSS score increase of ≥ 1.5 points if baseline EDSS = 0, ≥ 1.0 point if baseline EDSS <6.0, or ≥ 0.5 point if baseline EDSS ≥ 6.0 .¹⁹ All EDSS changes were confirmed at a second evaluation at least 3 months later. Evolution to a more severe stage of the disease was defined as CIS to RRMS, SPMS, or MS-related death; RRMS to SPMS or MS-related death; or SPMS to MS-related death. Transition from CIS to RRMS required the occurrence of a second clinical demyelinating event. SPMS was diagnosed retrospectively based on a history of gradual disability worsening following an initial relapsing disease course, with or without superimposed relapses, minor remissions, or plateaus.²⁵ MS-related death was identified through hospital records, local administrative databases, or proxy interviews, and classified according to previously published criteria.²⁰ Only deaths classified as MS-related were considered as outcome events in the analyses.

Neuropsychological testing was conducted using the Brief Repeatable Battery of Neuropsychological Tests (BRB-N)²⁹ at baseline and again at the 13-year and 26-year follow-ups. Tests assessed verbal memory (Selective Reminding Test), visuospatial memory (10/36 Spatial Recall), attention and information processing speed (Symbol Digit Modalities Test, Paced Auditory Serial Addition Test), and verbal fluency (Word List Generation). Z-scores for each BRB-N test were calculated using normative Italian data adjusted for age, sex, and education.^{30,31} To ensure consistency with the 13-year analysis,¹⁹ cognitive impairment was defined as failure in ≥ 2 BRB-N tests, with failure defined as a score ≥ 2 SD below the normative mean.³² Cognitive deterioration was defined as an increase in the number of failed neuropsychological tests at follow-up compared with baseline.³³ Long-term cognitive analyses were restricted to patients who were not cognitively impaired at baseline.

MRI Acquisition

Brain MRI scans were acquired at baseline and after 12 months (± 10 days) on a 1.5 T scanner (Magnetom SP63,

Siemens, Erlangen, Germany). During each session, the following brain sequences were obtained: (a) dual-echo spin-echo (repetition time [TR] = 2,400 msec, echo times [TE] = 30/80 msec, matrix = 256×256 , field of view [FOV] = 250×250 mm, slice thickness = 5 mm, 24 contiguous interleaved axial slices); (b) T1-weighted spin-echo (TR = 768 msec, TE = 15 msec, matrix = 256×256 , FOV = 250×250 mm, slice thickness = 5 mm, and 24 contiguous interleaved axial slices); (c) 2-dimensional gradient-echo (TR = 600 msec, TE = 12 msec; flip angle 20° , matrix = 256×256 , FOV = 250×250 mm, slice thickness = 5 mm, 20 contiguous axial slices covering the central brain regions. Images were acquired with and without an off-resonance saturation pulse (1.5 kHz offset; Gaussian envelope duration 16.4 msec; bandwidth 250 Hz; peak amplitude = 3.4×10^{-6} T).^{17,18,26,34}

At follow-up, participants were repositioned using baseline scans as anatomical reference to ensure longitudinal consistency.

MRI Analysis

WM Lesions

T2-hyperintense and T1-hypointense WM lesion volumes (LVs) were identified by agreement between 2 experienced observers on hard copies of proton density-weighted and T1-weighted scans, both blinded to subject identity and scan order. T2-weighted images were reviewed to increase confidence in lesion identification. Once identified, total T2-hyperintense and T1-hypointense WM LVs were quantified using a semiautomated, computer-assisted local thresholding technique previously described.^{17,18,26}

Volumetric MRI Measures

Brain volumetric measures were obtained using Statistical Parametric Mapping (SPM2; Wellcome Trust Centre for Neuroimaging, London, the United Kingdom). After correction for image intensity inhomogeneities using a maximum entropy-based algorithm, proton density-weighted and T2-weighted images were segmented to generate probabilistic maps of GM, WM, and CSF for each voxel.^{18,34} Because T2-visible lesions may be misclassified during tissue segmentation, all lesion voxels were reassigned to the WM compartment with probability equal to one and removed from the GM and CSF maps before tissue volume computation. Intracranial volume was calculated as the sum of GM, WM, and CSF volumes. Brain parenchymal fraction (BPF), GM fraction (GMF), and WM fraction were then obtained by dividing the corresponding tissue volumes by intracranial volume.^{18,34} Thalamic volume was measured using region-of-interest-based segmentation and normalized to intracranial volume to derive thalamic fraction.^{18,34}

MTR MRI Measures

MTR maps were generated on a pixel-by-pixel basis from coregistered magnetization transfer-on and magnetization transfer-off images using the standard formula $MTR = (M_0 - M_s)/M_0 \times 100$. Extracranial tissue was removed using local

thresholding, and pixels with MTR values <10% were excluded to reduce CSF-related partial volume effects. Normalized MTR histograms were generated for T2-hyperintense WM lesions, NAWM, GM, and thalamus after exclusion of macroscopic lesions from the corresponding tissue compartments.^{17,18,26,34} From each histogram, mean MTR and histogram peak height were derived.^{17,18,26,34} For thalamic MTR analysis, MTR maps were spatially normalized and a CSF mask was applied to minimize partial volume contamination.³⁴

12-Month MRI Longitudinal Changes

Longitudinal changes in MRI-derived measures were quantified as the absolute difference between 12 months and baseline values for LVs, volumetric fractions, and MTR metrics.^{17,18,26,34}

Statistical Analysis

We aimed to investigate whether baseline demographic, clinical, and MRI variables, as well as 12-month MRI longitudinal changes were associated with 26-year outcomes. High-efficacy disease-modifying therapy (HE-DMT) escalation was included as an explanatory variable. Clinical outcomes included: (1) EDSS worsening, (2) evolution to a more severe disease stage, (3) MS-related death, and (4) cognitive deterioration. T2-hyperintense and T1-hypointense WM LVs were log-transformed. Their percentage changes over 12 months were expressed as log-ratio. The associations of each variable with long-term outcomes were quantified as odds ratios from follow-up duration-adjusted logistic regression models, using Firth's bias-reduced estimation approach to address small sample sizes and moderate class imbalance observed for some outcomes. Given the exploratory and nonconclusive nature of the univariate analysis, no correction for multiplicity was applied. For each outcome, multivariable associations were assessed using logistic regression models with LASSO regularization. The tuning parameter, which controls the amount of penalization and promotes variable selection, was optimized by minimizing the deviance using 5-fold cross-validation, repeated 50 times to enhance stability. Model discrimination was assessed using the area under the receiver operating characteristic curve (AUC). Bootstrap resampling (1,000 replicates) allowed to evaluate the stability of variable selection across resamples and to estimate optimism-corrected AUC values, thereby accounting for potential overfitting. Sensitivity analyses excluded (1) participants lost to follow-up and (2) patients with SPMS at baseline. Stratification by outcome class was applied during both cross-validation and bootstrap resampling to preserve class proportions across folds and resamples, helping to mitigate potential instability arising from class imbalance. All univariable and multivariable analyses included follow-up duration and disease duration as covariates to adjust for potential confounding. R (v4.3.3) was used for computations.

Data Availability

The corresponding author had full access to all study data and takes responsibility for data integrity and analytical accuracy.

The anonymized dataset used for this study is available from the corresponding author on reasonable request.

Results

Baseline Characteristics and Evolution at 26-Year Follow-Up

The study included 73 patients from the original relapsing-onset MS cohort.^{17,18,26,34} Baseline characteristics and 12-month longitudinal changes in clinical and MRI measures are summarized in Table 1. At baseline, the mean age was 33.0 (SD = 9.3) years, 70% of patients were female, and median disease duration was 4.0 years (interquartile range [IQR] = 0.0–8.0). Clinical phenotypes included CIS (n = 20), RRMS (n = 34), and SPMS (n = 19). None of the patients were receiving DMT at baseline.

The median follow-up was 25.9 years (IQR = 18.1–27.1). At the 26-year follow-up, 37 patients were evaluated in-person and 25 by telephone interview (Figure 1).²⁸ Eleven patients (3 CIS, 4 RRMS, and 4 SPMS) did not participate in the 26-year follow-up visit due to unwillingness or loss to follow-up. For these patients, the most recent available follow-up clinical assessment was used (4.5-year¹⁷: n = 3; 8-year¹⁸: n = 1; 13-year¹⁹: n = 7). In the whole cohort, the median EDSS increased from 2.5 (IQR = 1.0–4.5) at baseline to 6.0 (IQR = 2.5–9.0) at 26 years. Fifty-eight patients (79.5%) experienced EDSS worsening.

Among 20 CIS patients, 17 converted to RRMS (5 of whom further progressed to SPMS), 2 died between years 8–13, and 1 remained CIS. Of the 34 RRMS patients, 18 transitioned to SPMS, 4 died between years 13–26, and 13 remained RRMS. Among the 19 SPMS patients, 12 died (7 between years 8–13 and 5 between years 13–26), and 7 remained SPMS. Consequently, by 26 years, 53 (72.6%) had progressed to a more severe disease stage.

A total of 18 patients (24.7%) died from MS-related causes: 7 between years 8–13, and 11 between years 13–26. MS-related causes of death included pulmonary infections (n = 7), death due to MS in patients with very high EDSS scores in the absence of alternative acute causes (n = 5), sepsis (n = 3), pulmonary embolism (n = 1), brainstem failure (n = 1), and suicide (n = 1). EDSS was set to 10 for these individuals. Because MS-related death was a predefined outcome, patients who died during follow-up were considered as having reached the 26-year outcome for EDSS worsening, evolution to a more severe disease stage, and MS-related death, even if they were not physically evaluated at the 26-year visit. No deaths unrelated to MS occurred during the follow-up.

Forty-eight patients underwent baseline neuropsychological evaluation; 37 underwent neuropsychological evaluation at 26-year follow-up, whereas 11 of them had neuropsychological data available from the 13-year evaluation.¹⁹ At

Table 1 Demographic, Clinical, and MRI Variables at Baseline and 12-Month Longitudinal Changes

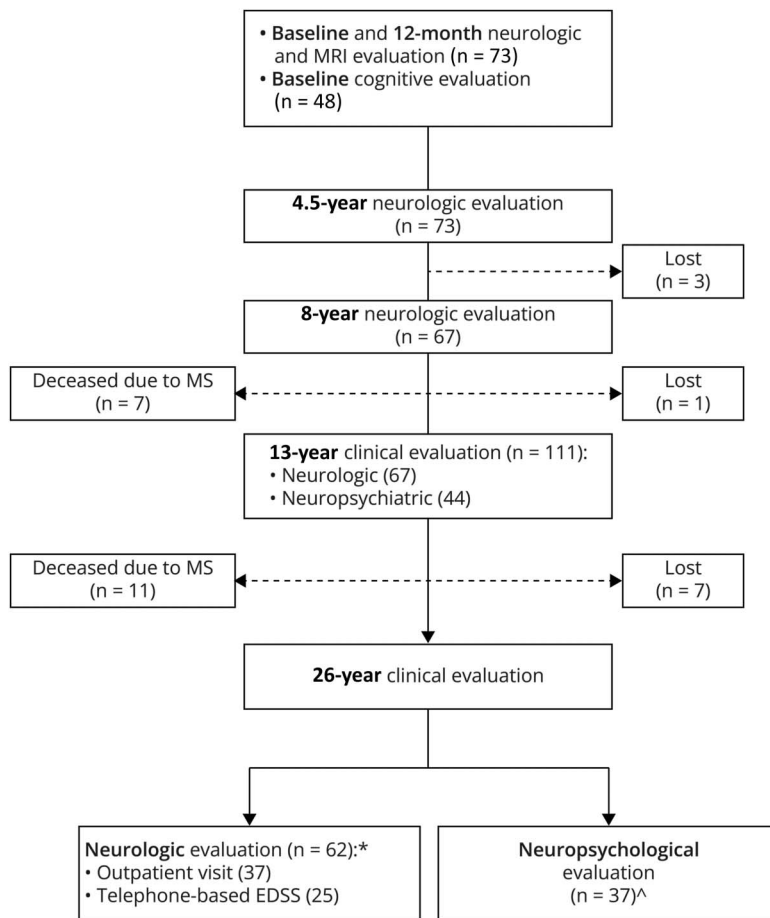
Variable	N = 73
Baseline age [y]	33.0 (9.3)
Sex, No. (%) Female/male	51 (69.9)/22 (30.1)
Baseline disease duration [y]	4.0 (0.0 to 8.0)
Baseline clinical phenotype, No. (%) CIS/RR/SP	20 (27.4)/34 (46.6)/19 (26.0)
Patients receiving DMT ^a at baseline, No. (%)	0 (0)
Patients starting DMT ^a during follow-up, No. (%)	73 (100)
Patients escalating to HE-DMT ^b during follow-up, No. (%)	31 (42.5)
Baseline EDSS score	2.5 (1.0 to 4.5)
12-mo EDSS score change	0.0 (0.0 to 0.5)
Baseline T2-hyperintense WM LV [mL]	12.0 (6.0 to 18.1)
Baseline T1-hypointense WM LV [mL]	0.4 (0.2 to 2.3)
Baseline BPF (%)	81.2 (78.8 to 82.7)
Baseline GMF (%)	46.5 (44.1 to 48.4)
Baseline WMF (%)	34.0 (32.3 to 36.4)
Baseline thalamic fraction (%)	5.7 (4.7 to 6.5)
Baseline T2-hyperintense WM lesion mean MTR	42.4 (40.5 to 44.7)
Baseline mean GM MTR	44.2 (42.3 to 46.1)
Baseline GM MTR histogram peak height	8.6 (6.4 to 9.8)
Baseline mean NAWM MTR	49.3 (48.5 to 50.2)
Baseline NAWM MTR histogram peak height	13.9 (12.2 to 15.5)
Baseline mean thalamic MTR	46.6 (45.3 to 48.9)
12-mo T2-hyperintense WM LV change (%)	6.6 (−16.7 to 31.2)
12-mo T1-hypointense WM LV change (%)	90.9 (13.3 to 433.7)
12-mo BPF change (%)	−0.4 (−1.6 to 1.2)
12-mo GMF change (%)	−0.6 (−4.3 to 2.0)
12-mo WMF change (%)	0.6 (−2.1 to 3.2)
12-mo thalamic fraction change (%)	−5.2 (−10.5 to 0.2)
12-mo mean T2-hyperintense WM lesion MTR change (%)	−3.5 (−6.6 to 1.4)
12-mo mean GM MTR change (%)	−4.0 (−6.9 to −0.9)
12-mo GM MTR histogram peak height change (%)	−1.8 (−7.2 to 5.6)
12-mo mean NAWM MTR change (%)	−3.9 (−5.5 to −1.7)
12-mo NAWM MTR histogram peak height change (%)	−2.2 (−9.3 to 6.0)
12-mo mean thalamic MTR change (%)	−0.2 (−5.6 to 2.3)

Abbreviations: BPF = brain parenchymal fraction; CIS = clinically isolated syndrome; DMT = disease-modifying therapy; EDSS = Expanded Disability Status Scale; GM = gray matter; GMF = gray matter fraction; HE-DMT = high-efficacy disease-modifying therapy; LV = lesion volume; MTR = magnetization transfer ratio; NAWM = normal-appearing white matter; RR = relapsing-remitting; SP = secondary progressive; WM = white matter; WMF = white matter fraction. Measures are reported as medians and interquartile ranges (25th–75th percentiles); age is reported as mean and standard deviation; sex, clinical phenotype, patients receiving DMT at baseline, patients starting DMT during follow-up and escalating to HE-DMT during follow-up are reported as frequencies and percentages.

^a DMT = interferon beta-1a, glatiramer acetate, azathioprine, methotrexate, mitoxantrone, cyclophosphamide, fingolimod, siponimod, natalizumab, rituximab, or ocrelizumab.

^b HE-DMT = mitoxantrone, cyclophosphamide, fingolimod, siponimod, natalizumab, rituximab, or ocrelizumab.

Figure 1 Study Flowchart



Flowchart showing the longitudinal description of the MS cohort under investigation. Counts at each visit denote actual assessment. Losses refer to nonparticipation at 26-year evaluation. Nonadditive totals reflect skipped visits. See text for further details. *Eleven patients of the original cohort¹⁷⁻¹⁹ did not undertake the 26-year outpatient visit (unwilling to attend or unreachable). For 3 of them, the EDSS scores and clinical status as assessed at the 4.5-year visit¹⁷ were entered in the analysis, for one, the 8-year evaluation¹⁸ was used, whereas for 7 was assessed at 13-year evaluation.¹⁹ During the follow-up, 18 patients had died before the 26-year evaluation due to MS-related causes, 7 of them between the 8- and 13-year evaluations, and 11 between the 13- and 26-year evaluations. **Patients with MS-related deaths (n = 18) were considered as having reached the 26-year outcomes for EDSS worsening, evolution to a more severe disease stage and MS-related death. These 18 deceased patients were included among those evaluated by telephone (n = 25), performed by interview with their caregivers. Forty-eight patients underwent an initial baseline neuropsychological evaluation. Then, 37 underwent the neuropsychological evaluation also at 26-year follow-up, whereas for 11 of them the 13-year neuropsychological evaluation¹⁹ was available and was used. Of note, 6 of these 11 patients died between years 13 and 26. However, because cognitive deterioration was defined as a worsening relative to baseline and had already occurred by the 13-year evaluation in these 6 patients, subsequent MS-related death did not affect the cognitive outcome classification. EDSS = Expanded Disability Status Scale; MS = multiple sclerosis.

baseline, 5 of 48 patients (10.4%) already met criteria for cognitive impairment; 17 of 43 (39.5%) showed cognitive deterioration at follow-up. Six patients with neuropsychological data available at the 13-year evaluation died between years 13 and 26; however, cognitive deterioration had already occurred by the 13-year visit.

All patients initiated DMT during follow-up; 25 (34.2%) began treatment within the first year. In total, 31 patients (42.5%) escalated to HE-DMTs during the 26-year period.

Univariable Associations With Clinical Outcomes at 26 Years

Univariable analyses examining associations of demographic, clinical, and MRI variables associated with outcomes are presented in eTable 1.

EDSS Worsening at 26 Years

In the analysis including all patients, EDSS worsening was associated with escalation to HE-DMT (standardized- β = 0.059), lower baseline GMF (standardized- β = -0.561), lower baseline thalamic fraction (standardized- β = -0.115), lower baseline NAWM MTR histogram peak height

(standardized- β = -0.010), and higher 12-month decline in GM MTR histogram peak height (standardized- β = -0.115) (AUC = 0.843, 95% CI 0.744–0.941) (Table 2, Figure 2).

Bootstrap resampling showed the highest selection frequencies for baseline GMF (90.2%), escalation to HE-DMT (67.6%), 12-month GM MTR histogram peak height decline (58.3%), 12-month EDSS score change (56.8%), baseline thalamic fraction (55.4%), and baseline NAWM MTR histogram peak height (51.7%) (optimism-corrected AUC = 0.709, 95% CI 0.599–0.795) (Table 2).

In sensitivity analyses excluding patients lost to follow-up, EDSS worsening was associated with lower baseline GMF (standardized- β = -0.495) and higher decline in 12-month GM MTR histogram peak height (standardized- β = -0.111), but also with higher 12-month EDSS score change (standardized- β = 0.036), and higher baseline T2-hyperintense WM LV (standardized- β = 0.223) (AUC = 0.866, 95% CI 0.763–0.970). When excluding patients with SPMS at baseline, escalation to HE-DMT (standardized- β = 0.180), lower baseline GMF (standardized- β = -0.374), lower baseline thalamic fraction (standardized- β = -0.124), and lower

Table 2 Multivariable Associations of Demographic, Clinical, and MRI Variables With EDSS Worsening and Evolution to a More Severe Stage at 26-Year Follow-Up

EDSS worsening						
Predictor	All patients (58/73 [79.5%])		Without patients lost at FU (51/62 [82.3%])		Without SPMS patient at baseline (40/55 [72.7%])	
	Std. β	Percentage of selection (%)	Std. β	Percentage of selection (%)	Std. β	Percentage of selection (%)
Escalation to HE-DMT ^a	0.059	67.6		46.1	0.180	66.6
Baseline age		6.4		1.5		7.2
Sex (male vs female)		26.3		22.4		21.1
Baseline EDSS score		7.3		10.0		7.3
12-mo EDSS score change		56.8	0.036	58.4		35.6
Baseline T2-hyperintense WM LV		33.4	0.223	64.4		23.2
Baseline T1-hypointense WM LV		10.1		9.4		7.5
Baseline BPF (%)		4.8		2.8		10.3
Baseline GMF (%)	-0.561	90.2	-0.495	72.6	-0.374	76.2
Baseline WMF (%)		38.0		38.2		17.0
Baseline thalamic fraction (%)	-0.115	55.4		17.7	-0.124	53.3
Baseline T2-hyperintense WM lesion mean MTR		23.1		22.6		15.4
Baseline mean GM MTR		18.8		18.3		15.7
Baseline GM MTR histogram peak height		33.7		9.7		19.4
Baseline mean NAWM MTR		13.1		22.8		22.6
Baseline NAWM MTR histogram peak height	-0.010	51.7		31.6	-0.152	53.4
Baseline mean thalamic MTR		6.7		7.5		13.7
12-mo T2-hyperintense WM LV change		27.3		26.4		9.1
12-mo T1-hypointense WM LV change		16.4		14.6		12.2
12-mo BPF change (%)		17.1		5.1		15.1
12-mo GMF change (%)		21.9		41.9		22.9
12-mo WMF change (%)		17.5		39.4		14.3
12-mo thalamic fraction change (%)		22.5		20.9		17.7
12-mo mean T2-hyperintense WM lesion MTR change (%)		17.4		8.2		20.6
12-mo mean GM MTR change (%)		40.0		21.2		20.8
12-mo GM MTR histogram peak height change (%)	-0.115	58.3	-0.111	52.4		41.9
12-mo mean NAWM MTR change (%)		15.6		11.1		4.6
12-mo NAWM MTR histogram peak height change (%)		15.2		9.7		12.0
12-mo mean thalamic MTR change (%)		34.1		32.2		14.0
	AUC (95% CI) 0.843 (0.744–0.941)	AUC optimism-corrected (95% CI) 0.709 (0.599–0.795)	AUC (95% CI) 0.866 (0.763–0.970)	AUC optimism-corrected (95% CI) 0.746 (0.642–0.843)	AUC (95% CI) 0.835 (0.723–0.947)	AUC optimism-corrected (95% CI) 0.693 (0.582–0.810)
Evolution to a more severe stage						
Predictor	All patients (53/73 [72.6%])		Without patients lost at FU (45/62 [72.6%])		Without SPMS patient at baseline (41/55 [74.5%])	
	Std. β	Percentage of selection (%)	Std. β	Percentage of selection (%)	Std. β	Percentage of selection (%)
Escalation to HE-DMT ^a		80.7		61.5	0.239	70.8
Baseline age		24.3		5.3		16.5
Sex (male vs female)		75.4		47.7	0.202	54.8
Baseline EDSS score		10.2		6.7		10.7
12-mo EDSS score change	0.167	89.9		50.8	0.188	59.5

Continued

Table 2 Multivariable Associations of Demographic, Clinical, and MRI Variables With EDSS Worsening and Evolution to a More Severe Stage at 26-Year Follow-Up (*continued*)

EDSS worsening						
Predictor	All patients (58/73 [79.5%])		Without patients lost at FU (51/62 [82.3%])		Without SPMS patient at baseline (40/55 [72.7%])	
	Std. β	Percentage of selection (%)	Std. β	Percentage of selection (%)	Std. β	Percentage of selection (%)
Baseline T2-hyperintense WM LV		67.7		45.8	0.912	90.1
Baseline T1-hypointense WM LV		2.5		4.8		2.0
Baseline BPF (%)		9.9		7.6		2.1
Baseline GMF (%)	-0.151	75.7	-0.080	70.5	-0.409	73.4
Baseline WMF (%)		44.5		41.1		26.2
Baseline thalamic fraction (%)		33.9		11.3	-0.043	51.5
Baseline T2-hyperintense WM lesion mean MTR		31.2		29.0		17.0
Baseline mean GM MTR		3.0		2.5		4.2
Baseline GM MTR histogram peak height		23.2		15.3		3.3
Baseline mean NAWM MTR		33.5		23.0		11.8
Baseline NAWM MTR histogram peak height		65.3		52.8		38
Baseline mean thalamic MTR		6.9		2.7		16.4
12-mo T2-hyperintense WM LV change		54.6	0.138	84.0		13.0
12-mo T1-hypointense WM LV change		37.8		18.5		23.9
12-mo BPF change (%)		23.2		24.4		15.3
12-mo GMF change (%)		17.6		16.5		9.0
12-mo WMF change (%)		18.6		18.6		16.3
12-mo thalamic fraction change (%)		24.4		15.1		11.0
12-mo mean T2-hyperintense WM lesion MTR change (%)		30.9		13.8		8.6
12-mo mean GM MTR change (%)		12.0		10.8		7.6
12-mo GM MTR histogram peak height change (%)		24.3		21.1		13.0
12-mo mean NAWM MTR change (%)		22.5		16.7		4.8
12-mo NAWM MTR histogram peak height change (%)		42.9		49.2		17.5
12-mo mean thalamic MTR change (%)		49.2		36.0		17.1
	AUC (95% CI) 0.793 (0.670–0.916)	AUC optimism-corrected (95% CI) 0.676 (0.589–0.755)	AUC (95% CI) 0.796 (0.686–0.906)	AUC optimism-corrected (95% CI) 0.693 (0.604–0.780)	AUC (95% CI) 0.929 (0.846–1.000)	AUC optimism-corrected (95% CI) 0.830 (0.725–0.916)

Abbreviations: AUC = area under the curve; BPF = brain parenchymal fraction; EDSS = Expanded Disability Status Scale; FU = follow-up; GM = gray matter; GMF = gray matter fraction; HE-DMT = high-efficacy disease-modifying therapy; LASSO = least absolute shrinkage and selection operator; LV = lesion volume; MTR = magnetization transfer ratio; NAWM = normal-appearing white matter; SPMS = secondary progressive MS; Std. β = standardized-beta coefficient; WM = white matter; WMF = white matter fraction.

Results were derived from multivariable logistic regression models with LASSO regularization. The optimal value of the tuning parameter (λ), which controls the degree of penalization and promotes variable selection, was determined through cross-validation; only variables with nonzero coefficients were retained in the final models. Standardized- β coefficients are reported to allow comparability of effect sizes across variables within the multivariable framework. To assess the robustness of variable selection, the analysis was repeated across 1,000 bootstrap resamples using the same modeling framework, and the percentage of times each variable was selected across resamples is reported. High selection frequencies indicate that a variable was consistently retained across resamples, suggesting potential relevance in alternative samples and reflecting sampling variability in the variable-selection process rather than indicating a lack of association. Model discriminative performance is summarized by the apparent AUC in the original sample and by the bootstrap-based optimism-corrected AUC. LASSO regression results for 2 sensitivity analyses - (1) excluding patients lost to follow-up and (2) excluding patients with SPMS at baseline - are also provided. Follow-up duration and disease duration were included as unpenalized covariates in all LASSO models to adjust for potential confounding. See the main text for further methodological details.

For each analysis, the number of patients meeting the outcome over the total number of patients included in the analysis, with the corresponding percentage in brackets, is specified.

^a HE-DMT = mitoxantrone, cyclophosphamide, fingolimod, siponimod, natalizumab, rituximab, or ocrelizumab.

baseline NAWM MTR histogram peak height (standardized- $\beta = -0.152$) were retained (AUC = 0.835, 95% CI 0.723–0.947). The identified variables consistently showed high selection frequency across bootstrap resamples, supporting the stability of the associations (Table 2).

Evolution to a More Severe Stage at 26 Years

Among all patients, evolution to a more severe disease stage was associated with higher 12-month EDSS score change (standardized- $\beta = 0.167$) and lower baseline GMF (standardized- $\beta = -0.151$) (AUC = 0.793, 95% CI 0.670–0.916) (Table 2, Figure 2).

Bootstrap analyses identified the highest selection frequencies for 12-month EDSS score change (89.9%), escalation to HE-DMT (80.7%), baseline GMF (75.7%), male sex (75.4%), baseline T2-hyperintense WM LV (67.7%), baseline NAWM MTR histogram peak height (65.3%), and 12-month T2-hyperintense WM LV change (54.6%) (optimism-corrected AUC = 0.676, 95% CI 0.589–0.755) (Table 2).

In the sensitivity analysis excluding patients lost to follow-up, the retained variables included lower baseline GMF (standardized- $\beta = -0.080$) and higher 12-month T2-hyperintense WM LV accumulation (standardized $\beta = 0.138$) (AUC = 0.796, 95% CI 0.686–0.906). After excluding patients with SPMS at baseline, higher 12-month EDSS score change (standardized- $\beta = 0.188$) and lower baseline GMF (standardized- $\beta = -0.409$) were retained, together with escalation to HE-DMT (standardized- $\beta = 0.239$), male sex (standardized- $\beta = 0.202$), higher baseline T2-hyperintense WM LV (standardized- $\beta = 0.912$), and lower baseline thalamic fraction (standardized- $\beta = -0.043$) (AUC = 0.929, 95% CI 0.846–1.000). Bootstrap resampling supported the robustness of these findings (Table 2).

MS-Related Death at 26 Years

In the analysis including all patients, MS-related death was associated with male sex (standardized- $\beta = 0.063$), higher baseline EDSS score (standardized- $\beta = 0.182$), lower baseline GMF (standardized- $\beta = -0.304$), lower baseline GM MTR histogram peak height (standardized- $\beta = -0.050$), and higher 12-month mean thalamic MTR decline (standardized- $\beta = -0.370$) (AUC = 0.887, 95% CI 0.813–0.961) (Table 3, Figure 3).

In the bootstrap analysis, the variables with the highest selection frequencies were 12-month mean thalamic MTR decline (75.1%), male sex (70.0%), baseline GMF (69.0%), baseline EDSS score (66.3%), baseline age (59.9%), baseline thalamic fraction (52.4%), and baseline GM MTR histogram peak height (51.2%) (optimism-corrected AUC = 0.811, 95% CI 0.748–0.866) (Table 3).

In the sensitivity analysis excluding patients lost to follow-up, MS-related death was associated with lower baseline GMF (standardized- $\beta = -0.172$) and lower baseline GM MTR

histogram peak height (standardized- $\beta = -0.331$) (AUC = 0.963, 95% CI 0.922–1.000). Bootstrap resampling supported the stability of these associations (Table 3). Sensitivity analysis excluding patients with SPMS at baseline was not performed because of the limited number of events ($n = 6$).

Cognitive Deterioration at 26 Years

Among patients with available cognitive data ($n = 43$), cognitive deterioration was associated with higher baseline age (standardized- $\beta = 0.197$), lower baseline BPF (standardized- $\beta = -0.039$), and lower baseline mean GM MTR (standardized- $\beta = -0.376$) (AUC = 0.891, 95% CI 0.788–0.994) (Table 3, Figure 3).

In the bootstrap analysis, the variables with the highest selection frequencies were baseline age (65.3%) and baseline mean GM MTR (62.3%) (optimism-corrected AUC = 0.819, 95% CI 0.742–0.925) (Table 3).

In the sensitivity analysis excluding patients lost to follow-up, cognitive deterioration was associated with higher baseline age (standardized- $\beta = 0.120$), and baseline mean GM MTR (standardized- $\beta = -0.423$) but also with higher baseline T2-hyperintense WM LV (standardized- $\beta = 0.025$), and lower baseline mean NAWM MTR (standardized- $\beta = -0.050$) (AUC = 0.876, 95% CI 0.718–1.000). In the sensitivity analysis excluding patients with SPMS at baseline, cognitive deterioration was associated with higher baseline age (standardized- $\beta = 0.441$) and lower baseline mean GM MTR (standardized- $\beta = -0.630$), but also with higher baseline T2-hyperintense and T1-hypointense WM LVs (standardized- $\beta = 0.102$ and 0.038 , respectively) (AUC = 0.927, 95% CI 0.844–1.000). Bootstrap resampling provided insight into variable-selection stability across resamples, with some variability observed as expected given the more limited sample size (Table 3).

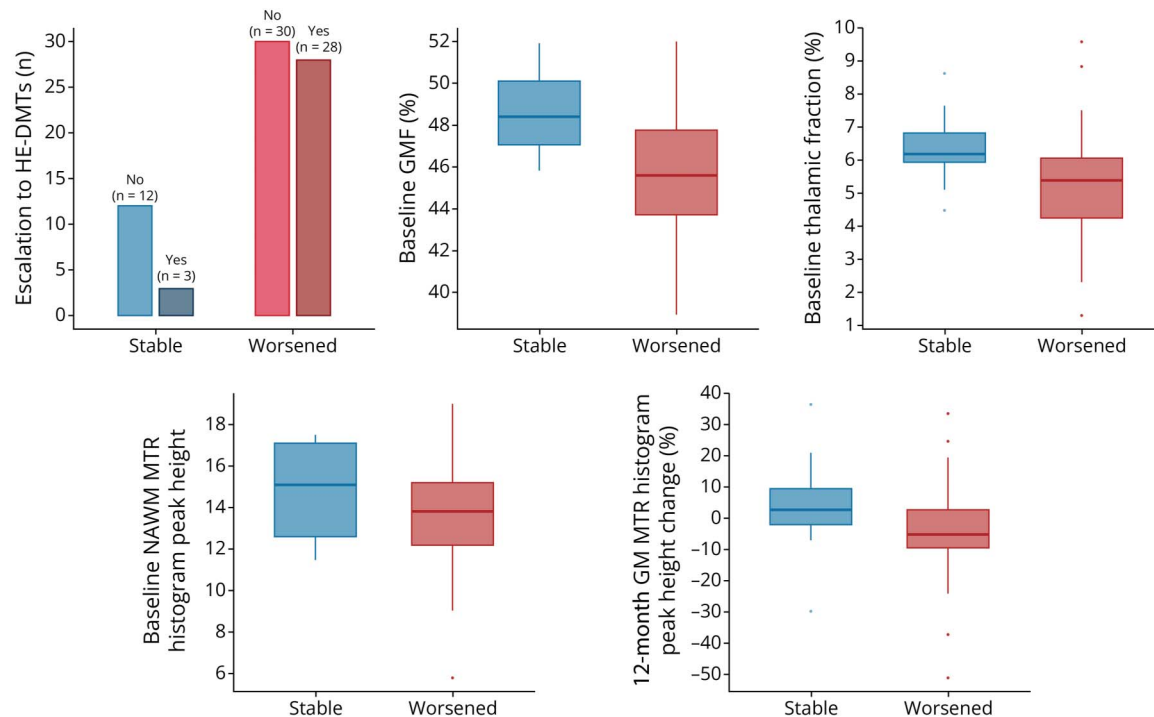
Discussion

GM damage, assessed through volumetric and MTR MRI measures, was consistently associated with multiple long-term clinical outcomes in relapse-onset MS over a 26-year follow-up. These associations were robust across analytical approaches. In contrast, WM-related measures showed less consistent associations, emerging primarily in relation to disease stage evolution and in sensitivity analyses.

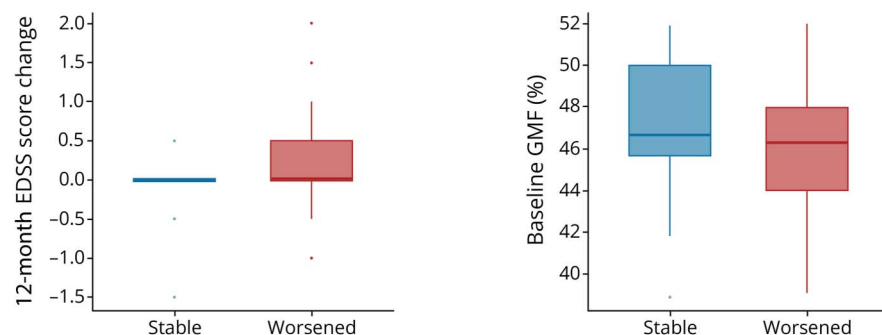
Over a 26-year follow-up, most CIS patients in our cohort developed clinically definite MS, consistent with epidemiologic data reporting 60%–80% conversion rates over 30 years.^{4,5,7,35} In line with reported SPMS conversion rates of 2%–3% annually,^{6,35,36} 37% of patients with CIS or RRMS at baseline progressed to SPMS, whereas EDSS worsening occurred in 79.5% of the cohort, comparable with rates observed in natural history studies conducted before the widespread adoption of HE-DMTs.^{6,14,35,36} MS-related death was

Figure 2 Distribution of Variables Associated With EDSS Worsening and Evolution to a More Severe Stage at 26-Year Follow-Up

A. EDSS worsening



B. Evolution to a more severe stage



The bar plot and box plots depict the distribution of demographic, clinical, and MRI variables retained in the LASSO-regularized logistic regression analysis including all patients, shown according to outcome status for (A) EDSS worsening and (B) evolution to a more severe disease stage. See text for further details. EDSS = Expanded Disability Status Scale; GM = gray matter; GMF = gray matter fraction; HE-DMT = high-efficacy disease-modifying therapy; LASSO = least absolute shrinkage and selection operator; MTR = magnetization transfer ratio; NAWM = normal appearing white matter.

recorded in 24.7% of the cohort, predominantly in SPMS patients, consistent with the known life expectancy reduction in MS and the higher mortality rate compared with the general population, particularly in the pre-DMT era.^{21,22,37} Cognitive deterioration affected 39.5% of patients who were cognitively preserved at baseline, in line with longitudinal studies reporting prevalence estimates of 28%–29% of cognitive decline in MS over 5 years.^{38,39}

EDSS worsening at 26 years was associated with escalation to HE-DMT, lower baseline GM fraction, lower baseline thalamic fraction, lower baseline NAWM MTR histogram peak height, and greater 12-month decline in GM MTR histogram

peak height. These associations remained consistent across penalized multivariable models, bootstrap resampling, and sensitivity analyses, extending earlier findings from this cohort at 4.5, 8, and 13 years.^{17–19,34} Our findings support emerging evidence that GM damage, both global and structure-specific, particularly involving the thalamus,^{6,34,40} captures pathologic processes that continue to influence disability accumulation decades later.^{6,12–15} Importantly, this association was evident from early disease phases, as supported by sensitivity analyses excluding patients with SPMS at baseline.

This progressive relevance of GM MRI metrics over time supports a temporal shift in the biological drivers of disability

Table 3 Multivariable Associations of Demographic, Clinical, and MRI Variables With MS-Related Death and Cognitive Deterioration at 26-Year Follow-Up

MS-related death						
Predictor	All patients (18/73 [24.7%])		Without patients lost at FU (18/62 [29.0%])			
	Std. β	Percentage of selection (%)	Std. β	Percentage of selection (%)		
Escalation to HE-DMT ^a		47.0		6.5		
Baseline age		59.9		10.2		
Sex (male vs female)	0.063	70.0		37.7		
Baseline EDSS score	0.182	66.3		31.3		
12-mo EDSS score change		56.2		24.8		
Baseline T2-hyperintense WM LV		7.7		0.9		
Baseline T1-hypointense WM LV		3.4		5.8		
Baseline BPF (%)		7.2		15.9		
Baseline GMF (%)	−0.304	69.0	−0.172	58.6		
Baseline WMF (%)		36.4		17.4		
Baseline thalamic fraction (%)		52.4		51.6		
Baseline T2-hyperintense WM lesion mean MTR		15.0		0.8		
Baseline mean GM MTR		6.7		6.6		
Baseline GM MTR histogram peak height	−0.050	51.2	−0.331	66.1		
Baseline mean NAWM MTR		5.3		1.0		
Baseline NAWM MTR histogram peak height		21.5		3.9		
Baseline mean thalamic MTR		16.2		2.9		
12-mo T2-hyperintense WM LV change		46.0		6.9		
12-mo T1-hypointense WM LV change		7.7		2.8		
12-mo BPF change (%)		11.8		8.7		
12-mo GMF change (%)		15.3		16.2		
12-mo WMF change (%)		9.8		11.4		
12-mo thalamic fraction change (%)		17.4		17.4		
12-mo mean T2-hyperintense WM lesion MTR change (%)		19.6		2.1		
12-mo mean GM MTR change (%)		18.7		27.0		
12-mo GM MTR histogram peak height change (%)		17.3		0.1		
12-mo mean NAWM MTR change (%)		17.3		23.1		
12-mo NAWM MTR histogram peak height change (%)		27.3		5.7		
12-mo mean thalamic MTR change (%)	−0.370	75.1		26.1		
	AUC (95% CI) 0.887 (0.813–0.961)	AUC optimism-corrected (95% CI) 0.811 (0.748–0.866)	AUC (95% CI) 0.963 (0.922–1.000)	AUC optimism-corrected (95% CI) 0.931 (0.855–0.962)		
Cognitive deterioration						
Predictor	All patients (17/43 [39.5%])		Without patients lost at FU (14/33 [42.4%])		Without SPMS patient at baseline (12/36 [33.3%])	
	Std. β	Percentage of selection (%)	Std. β	Percentage of selection (%)	Std. β	Percentage of selection (%)
Escalation to HE-DMT ^a		19.8		8.8		16.6
Baseline age	0.197	65.3	0.120	51.3	0.441	75.8
Sex (male vs female)		43.7		15.7		36.2
Baseline EDSS score		10.1		4.9		32.3
12-mo EDSS score change		15.1		8.7		20.2

Continued

Table 3 Multivariable Associations of Demographic, Clinical, and MRI Variables With MS-Related Death and Cognitive Deterioration at 26-Year Follow-Up (*continued*)

MS-related death						
Predictor	All patients (18/73 [24.7%])		Without patients lost at FU (18/62 [29.0%])			
	Std. β	Percentage of selection (%)	Std. β	Percentage of selection (%)		
Baseline T2-hyperintense WM LV		36.2	0.025	34.2	0.102	54.5
Baseline T1-hypointense WM LV		9.9		14.6	0.038	35.4
Baseline BPF (%)	-0.039	41.0		26.8		29.9
Baseline GMF (%)		24.0		18.8		20.5
Baseline WMF (%)		2.7		4.3		3.8
Baseline thalamic fraction (%)		15.7		18.0		14.7
Baseline T2-hyperintense WM lesion mean MTR		10.2		7.4		6.6
Baseline mean GM MTR	-0.376	62.3	-0.423	66.3	-0.630	73.8
Baseline GM MTR histogram peak height		18.7		4.4		9.2
Baseline mean NAWM MTR		10.4	-0.050	22.9		22.0
Baseline NAWM MTR histogram peak height		4.9		9.0		2.2
Baseline mean thalamic MTR		5.7		0.4		10.3
12-mo T2-hyperintense WM LV change		13.9		16.7		13.6
12-mo T1-hypointense WM LV change		16.8		7.5		12.2
12-mo BPF change (%)		28.5		25.1		15.8
12-mo GMF change (%)		6.9		2.4		9.7
12-mo WMF change (%)		6.7		7.2		4.8
12-mo thalamic fraction change (%)		15.7		13.9		19.0
12-mo mean T2-hyperintense WM lesion MTR change (%)		16.3		20.1		19.7
12-mo mean GM MTR change (%)		1.9		0.8		1.6
12-mo GM MTR histogram peak height change (%)		1.2		1.9		2.6
12-mo mean NAWM MTR change (%)		1.4		2.1		1.2
12-mo NAWM MTR histogram peak height change (%)		13.0		4.5		15.3
12-mo mean thalamic MTR change (%)		18.2		10.3		4.7
	AUC (95% CI) 0.891 (0.788–0.994)	AUC optimism-corrected (95% CI) 0.819 (0.742–0.925)	AUC (95% CI) 0.876 (0.718–1.000)	AUC optimism-corrected (95% CI) 0.812 (0.699–0.978)	AUC (95% CI) 0.927 (0.844–1.000)	AUC optimism-corrected (95% CI) 0.846 (0.733–0.924)

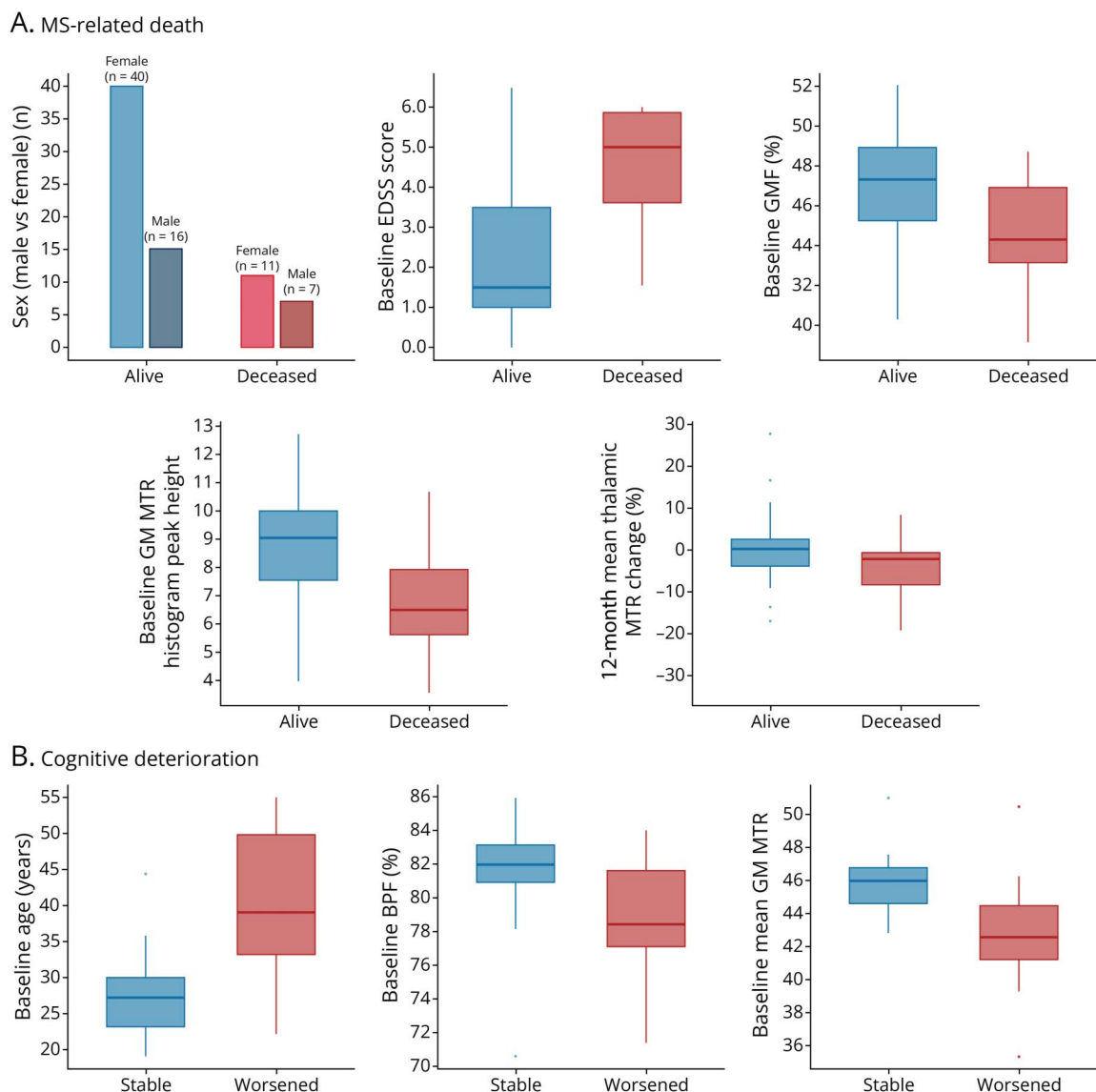
Abbreviations: AUC = area under the curve; BPF = brain parenchymal fraction; EDSS = Expanded Disability Status Scale; FU = follow-up; GM = gray matter; GMF = gray matter fraction; HE-DMT = high-efficacy disease-modifying therapy; LASSO = least absolute shrinkage and selection operator; LV = lesion volume; MS = multiple sclerosis; MTR = magnetization transfer ratio; NAWM = normal-appearing white matter; SPMS = secondary progressive MS; Std. β = standardized-beta coefficient; WM = white matter; WMF = white matter fraction.

Results were derived from multivariable logistic regression models with LASSO regularization. The optimal value of the tuning parameter (λ), which controls the degree of penalization and promotes variable selection, was determined through cross-validation; only variables with nonzero coefficients were retained in the final models. Standardized- β coefficients are reported to allow comparability of effect sizes across variables within the multivariable framework. To assess the robustness of variable selection, the analysis was repeated across 1,000 bootstrap resamples using the same modeling framework, and the percentage of times each variable was selected across resamples is reported. High selection frequencies indicate that a variable was consistently retained across resamples and therefore suggest greater stability of the association. Conversely, low selection frequencies indicate that the variable was rarely retained across resamples, which is consistent with its exclusion in the model fitted on the original sample and suggests limited relevance for the outcome. Notably, variables with coefficients equal to zero in the model fitted on the original sample may still appear in bootstrap resamples with non-negligible selection frequencies, suggesting potential relevance in alternative samples and reflecting sampling variability in the variable-selection process rather than indicating a lack of association. Model discriminative performance is summarized by the apparent AUC in the original sample and by the bootstrap-based optimism-corrected AUC. LASSO regression results for 2 sensitivity analyses - (1) excluding patients lost to follow-up and (2) excluding patients with SPMS at baseline - are also provided. MS-related death occurred in only 6 patients in the sensitivity analysis excluding SPMS patients; therefore, no association analyses were performed for this outcome. Follow-up duration and disease duration were included as unpenalized covariates in all LASSO models to adjust for potential confounding. See the main text for further methodological details.

For each analysis, the number of patients meeting the outcome over the total number of patients included in the analysis, with the corresponding percentage in brackets, is specified.

^a HE-DMT = mitoxantrone, cyclophosphamide, fingolimod, siponimod, natalizumab, rituximab, or ocrelizumab.

Figure 3 Distribution of Variables Associated With MS-Related Death and Cognitive Deterioration at 26-Year Follow-Up



The bar plot and box plots depict the distribution of demographic, clinical, and MRI variables retained in the LASSO-regularized logistic regression analysis including all patients, shown according to outcome status for (A) MS-related death and (B) cognitive deterioration. See text for further details. BPF = brain parenchymal fraction; EDSS = Expanded Disability Status Scale; GM = gray matter; GMF = gray matter fraction; LASSO = least absolute shrinkage and selection operator; MS = multiple sclerosis; MTR = magnetization transfer ratio.

in MS. While WM LVs and NAWM microstructural abnormalities were more relevant in shorter follow-up periods¹⁷ and in earliest phases of the disease,^{5,7} their associations diminished with longer observations and in patients with long standing MS, whereas GM measures became increasingly dominant. MRI measures of brain GM damage were associated with disability worsening by year 8,¹⁸ became more consistent by 13 years,¹⁹ and were clearly dominant by year 26. This pattern is consistent with a shift from inflammatory demyelination to neurodegenerative processes affecting brain GM and being more closely linked to irreversible disability.^{6,35}

Evolution to a more severe disease stage was associated with higher 12-month EDSS score change and lower baseline

GMF, indicating that both early disability worsening and lower GM volume, reflecting substantial neuro-axonal loss, are relevant for long-term disease evolution. Bootstrap analyses highlighted a broader set of variables that likely reflects the biological heterogeneity inherent in disease stage transitions, which include conversion from CIS to RRMS, progression to SPMS, and MS-related death, each potentially driven by partially distinct mechanisms.^{6,18,19,41} Sensitivity analyses further supported this interpretation. After excluding patients lost to follow-up, lower baseline GMF and higher 12-month WM lesion accumulation were retained. Exclusion of patients with SPMS at baseline revealed associations with both GM-related measures and WM LVs, along with male sex and HE-DMT escalation. Overall, these findings suggest that

GM damage represents a core substrate of long-term disease evolution, while WM inflammatory activity remains relevant for stage transitions earlier in the disease course, consistent with CIS studies.^{5,7}

Older age, progressive MS phenotype, motor and sensory involvement at clinical onset, higher EDSS scores, neurogenic bladder, pressure ulcers, and comorbidities have been associated with increased mortality risk.³⁷ However, the associations of MRI features with MS-related mortality remains incompletely characterized. In a prior 21-year longitudinal study, higher baseline T2-hyperintense WM LVs was associated with an increased mortality risk in interferon beta-1b-treated RRMS patients.²¹ In our study, male sex, higher baseline EDSS score, lower baseline GMF, lower baseline GM MTR histogram peak height, and greater 12-month decline in thalamic MTR were associated with long-term MS-related mortality, highlighting the importance of GM neurodegeneration in long-term survival. Bootstrap and sensitivity analyses excluding patients lost to follow-up reinforced this interpretation.

Although MS-related death was considered the terminal stage of disease evolution, mortality is also influenced by aging and comorbidities, which could not be fully accounted for in this long-term observational study. Moreover, although MS-related death was classified according to published criteria,²⁰ attribution of cause of death in long-term observational studies remains inherently complex, and some degree of misclassification cannot be entirely excluded.

Among patients cognitively preserved at baseline, cognitive deterioration was associated with older age, lower baseline BPF, and lower baseline mean GM MTR, indicating that age-related vulnerability, brain irreversible tissue loss, and GM microstructural damage may be associated with long-term cognitive decline.^{19,38,42}

Bootstrap analyses identified baseline age and baseline mean GM MTR as the only variables retained in more than 50% of resamples, supporting the robustness of these associations. Sensitivity analyses consistently retained baseline age and baseline GM MTR, whereas WM LVs and NAWM MTR measures emerged only in these analyses. Accordingly, WM damage may contribute to cognitive outcomes in earlier disease phases, possibly through disconnection mechanisms,⁴³ while not representing the dominant substrate of long-term cognitive deterioration. Cognitive deterioration was defined as a worsening relative to baseline performance, based on an increase in the number of failed neuropsychological tests; using this definition, 39.5% of patients being cognitively preserved at baseline were classified as having cognitive deterioration at 26 years. Although alternative criteria might identify a higher proportion of patients as deteriorated, this definition was deliberately retained to ensure methodological consistency with the previous 13-year follow-up analysis of the same cohort.¹⁹

Escalation to HE-DMT was associated with EDSS worsening and evolution to a more severe disease stage in sensitivity analysis excluding patients being SPMS at baseline. However, these association should be interpreted with caution. In patients with RRMS, escalation often reflects high inflammatory activity or insufficient disease control and may therefore be associated with a more aggressive disease course. By contrast, in patients with SPMS, escalation frequently coincided with the availability of HE-DMTs as the only approved treatment options, rather than reflecting a change in underlying disease activity or worsening. Accordingly, escalation to HE-DMTs should not be interpreted as a direct measure of disease worsening or as a causal determinant of long-term outcomes. Rather, it represents a context-dependent indicator influenced by disease phenotype, therapeutic era, and treatment availability and is subject to confounding by indication. Moreover, treatment selection was not randomized and evolved substantially over the long observation period, and escalation provides only a simplified summary of treatment exposure without capturing timing, sequencing, duration, or cumulative effects. In addition, HE-DMTs were largely unavailable during the early disease years for most patients, limiting the prevention of early irreversible CNS damage.¹⁶

Several limitations warrant consideration. A concomitant healthy control group was not included in the present analyses; however, healthy controls were examined in previous studies using identical MRI protocols to assess longitudinal stability of these measures and disease-related abnormalities.^{17,18,26,34} The present work specifically focused on the associations of these MRI measures with long-term outcomes in MS. Clinical and MRI assessments were performed at few predefined intervals, precluding time-dependent analyses for clinical outcomes and treatment changes and differentiation between relapse-associated worsening and progression independent of relapse activity. Telephone-based EDSS assessment does not allow a full neurologic examination and may be less reliable at lower disability levels. However, it has been validated against in-person assessment, shows high agreement at higher EDSS scores, and is widely used in long-term studies, particularly for patients with severe disability or limited access to MS centers.²⁸ The lack of systematically collected comorbidity data represents another limitation, as comorbid conditions may negatively influence long-term outcomes. Cognitive impairment was defined as failure in ≥ 2 BRB-N tests regardless of cognitive domain; despite its limitations, this criterion was widely adopted at the time of study design and retained for consistency with earlier follow-up analyses of the same cohort. Additional limitations relate to MRI acquisition and analysis. The spatial resolution and the use of 2-dimensional sequences may have introduced partial volume effects, particularly affecting GM-derived volumetric and MTR measures, despite masking and thresholding strategies. The lack of continuous follow-up imaging precluded dynamic assessment of lesion activity, atrophy rates, or DMT effects. Heterogeneity arising from inclusion of patients at

different disease stages may limit generalizability, and stratified analyses in larger, more homogeneous cohorts are needed. The absence of spinal cord imaging, reflecting historical MRI protocols, represents another important limitation. Because spinal cord pathology contributes to disability accumulation in MS,⁴⁴⁻⁴⁶ its absence may represent an unmeasured confounder to long-term disability progression and disease stage evolution. Nonetheless, the consistency of our findings with previous studies supports brain GM damage as a relevant substrate of disease severity and progression. Moreover, while our protocol incorporated advanced MRI techniques for its time, it lacked newer modalities included those for detecting cortical^{6,47,48} and chronic active lesions,^{6,46} which are associated with worse disease evolution. Future research should integrate brain and spinal cord imaging and incorporate novel MRI markers to capture the full spectrum of MS pathology.

In conclusion, GM damage is consistently associated with clinical outcomes over more than 2 decades. These findings highlight the potential relevance of GM-sensitive MRI metrics for understanding long-term disease evolution and underscore the need for further validation in larger, independent cohorts, using modern imaging protocols and contemporary treatment strategies before considering their integration into prognostic or clinical frameworks.

Author Contributions

P. Preziosa: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data. A. Meani: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. N. Tedone: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. I. Gattuso: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data. M. Rovaris: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; analysis or interpretation of data. M.A. Rocca: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. M. Filippi: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data.

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